

RieszNet and ForestRiesz: Automatic Debiased Machine Learning with Neural Nets and Random Forests

Victor Chernozhukov; Whitney Newey; Víctor Quintas-Martínez; Vasilis Syrgkanis

01

RieszNet and ForestRiesz learn α to align plug-in and DR

- Targets: linear functionals (ATE, policy value, average derivatives); naive ML plug-ins are biased.
- Learn the Riesz representer α to form a doubly robust score that cancels first-order bias.
- Neyman orthogonality + cross-fitting yield root-n inference when the nuisance MSE product is small.
- RieszNet (NN) and ForestRiesz (GRF) learn α via the Riesz loss and a brief targeted update; cross-fitted joint training yields valid root-n inference.

02 IHDP and BHP are used only to evaluate methods

- IHDP: used to evaluate ATE estimators.
- BHP: used to evaluate average derivative estimators.

03

Assumptions and design choices constrain performance—and open questions remain

- RR existence and MSE continuity can fail under weak overlap or heavy tails; cross-fitting design (folds/splits) affects robustness; double cross-fitting can help.
- Targeted update ϵ can inflate variance; use regularization, clipping, and diagnostics.
- Choosing the α -class: smoothness vs. sparsity; diagnostics for misspecification.
- Weak-signal and finite-sample fragility: when $\hat{\alpha}$ and $\hat{g}$ are noisy or error-product $\approx n^{-1/2}$; use robust variance estimation and safeguards.
- Computation and scalability: optimization stability, tuning robustness, reproducibility.
- Extensions to explore: policy value, off-policy, IV, missingness, sequential treatments.
- When to prefer RieszNet vs. ForestRiesz; known failure modes.
- Theory gap: finite-sample guarantees for learned α under weak overlap

Thank You

Questions & Discussion